Prabhavit Gupta

Artificial Intelligence Engineer | LLMs, Transformers, Model Optimization, RAG & MLOps

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Executive Summary

- **AI/ML Engineering & Open-Weight LLM Deployment:** AI Engineer with 3 years commercial software and applied machine learning experience. Skilled in deploying, adapting, and optimizing transformer-based architectures and open-weight Large Language Models (LLaMA 3, Mistral, Falcon) alongside commercial APIs for domain-specific business automation.
- **Model Optimization, Quantization & Low-Latency Serving:** Hands-on experience optimizing model inference using modern acceleration techniques (4-bit/8-bit quantization via AWQ/GPTQ, vLLM high-throughput serving runtimes, prompt distillation, and caching strategies), drastically reducing compute memory footprints and per-query latency.
- **Production RAG Pipelines, Vector Search & MLOps:** Proven background architecting Retrieval-Augmented Generation (RAG) frameworks using LangChain, LlamaIndex, and vector databases (pgvector, Qdrant). Experienced implementing automated data ingestion pipelines, containerized deployment (Docker, Kubernetes), and model observability workflows.

Technical Skills

- **Core Machine Learning & LLMs:** Python, PyTorch, Hugging Face Transformers, LLaMA 3, Mistral, Ollama, vLLM, OpenAI API, Anthropic Claude API, Parameter-Efficient Adaptation (LoRA/QLoRA principles).
- **Model Optimization & Serving:** Quantization (AWQ, GPTQ, bitsandbytes), KV-Cache Optimization, Semantic Caching (Redis), Latency/Throughput Profiling, Token Budget Management.
- **RAG Architectures & Vector Stores:** LangChain, LlamaIndex, LangGraph, Vector Databases (PostgreSQL pgvector, Qdrant, ChromaDB), Hybrid Search (BM25 + Dense Embeddings), Cross-Encoder Reranking.
- **Data Pipelines & Backend:** Data Ingestion for Unstructured Documents, FastAPI, AsyncIO, PyDantic, SQL, REST APIs, JSON Schema Enforcement.
- **MLOps & Cloud Infrastructure:** Docker, Docker Compose, Git, GitHub Actions CI/CD, Linux Server Environments, AWS (S3, Lambda, EC2), Model Monitoring & Drift Tracking.

Professional Experience

AI/ML Engineer & Solutions Architect • *Independent / Applied AI Consultancy*
2024 – Present • Venice, IT (Remote Italy)

- Designed and deployed end-to-end AI applications combining open-source transformer models (Mistral 7B, LLaMA 3) hosted via vLLM with proprietary LLM APIs for automated enterprise document analysis.
- Implemented model quantization (AWQ 4-bit) and asynchronous batch inference pipelines, reducing server memory utilization by 60% and improving inference throughput by 2.8x.
- Architected high-accuracy RAG systems over multi-thousand page technical datasets, integrating semantic chunking, embedding generation, pgvector indexing, and reranking to eliminate hallucinations.
- Developed robust data preprocessing and validation pipelines in Python to clean, structure, and tokenize proprietary datasets for domain-specific model fine-tuning and retrieval.
- Engineered RESTful microservices with FastAPI and containerized workloads in Docker, implementing health-check endpoints, automated regression testing, and CI/CD deployment pipelines.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Developed backend data processing services in Python and TypeScript, integrating machine learning classifiers and NLP models into business-critical workflows.
- Managed and scaled PostgreSQL databases, optimizing vector similarity searches and indexing strategies for high-concurrency read operations.
- Implemented asynchronous task orchestration using message queues, managing distributed data pipelines and background ingestion jobs reliably.
- Collaborated with product teams to translate domain problems into technical AI features, conducting evaluation benchmarks and monitoring model accuracy in production.

Key Engineering Projects

- **High-Throughput Open-Source LLM Serving Pipeline:** Deployed containerized vLLM inference server running quantized Mistral models with dynamic batching, exposed through a FastAPI proxy with Redis semantic caching and token-usage telemetry.

- **Enterprise Semantic Knowledge Engine (Python + pgvector):** Built an intelligent document retrieval platform supporting hybrid search, custom metadata filtering, and automated evaluation benchmarking via RAGAS metrics.

Education & Languages

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, mathematical modeling, decision analysis, and quantitative methods.
- **Languages:** English (C1 Professional, fluent written/spoken), Italian (Working proficiency / fluent professional communication).
- **Legal & Work Authorization:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to work across Italy without visa sponsorship; Registered Italian Partita IVA ready.